Task 2

(1)

```
## Call:
## ergm(formula = glasgow_net1 ~ edges + nodematch("gender"))
##
## Maximum Likelihood Results:
##
##                  Estimate Std. Error MCMC % z value Pr(>|z|)
## edges             -5.0133     0.1365      0  -36.72   <1e-04 ***
## nodematch.gender   2.0098     0.1460      0   13.77   <1e-04 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      Null Deviance: 22890  on 16512  degrees of freedom
##  Residual Deviance:  3824  on 16510  degrees of freedom
##
## AIC: 3828  BIC: 3843  (Smaller is better. MC Std. Err. = 0)
```

- Negative (and significant) edge coefficient - networks tend to be sparse
- Positive (and significant) homophily coefficient - networks tend to have homophily

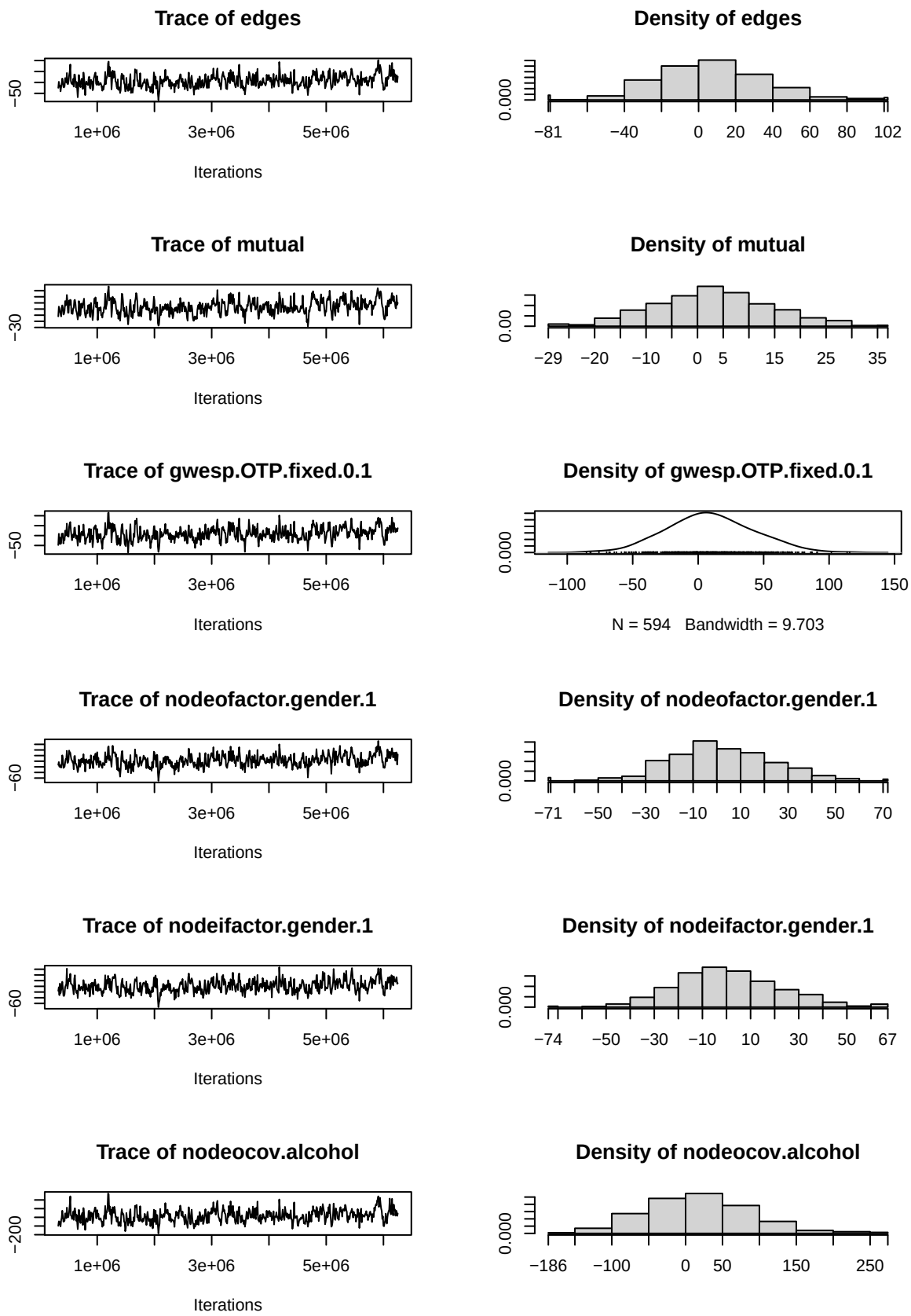
```
## Conditional probability of a tie between same-gender students: 0.04726488
```

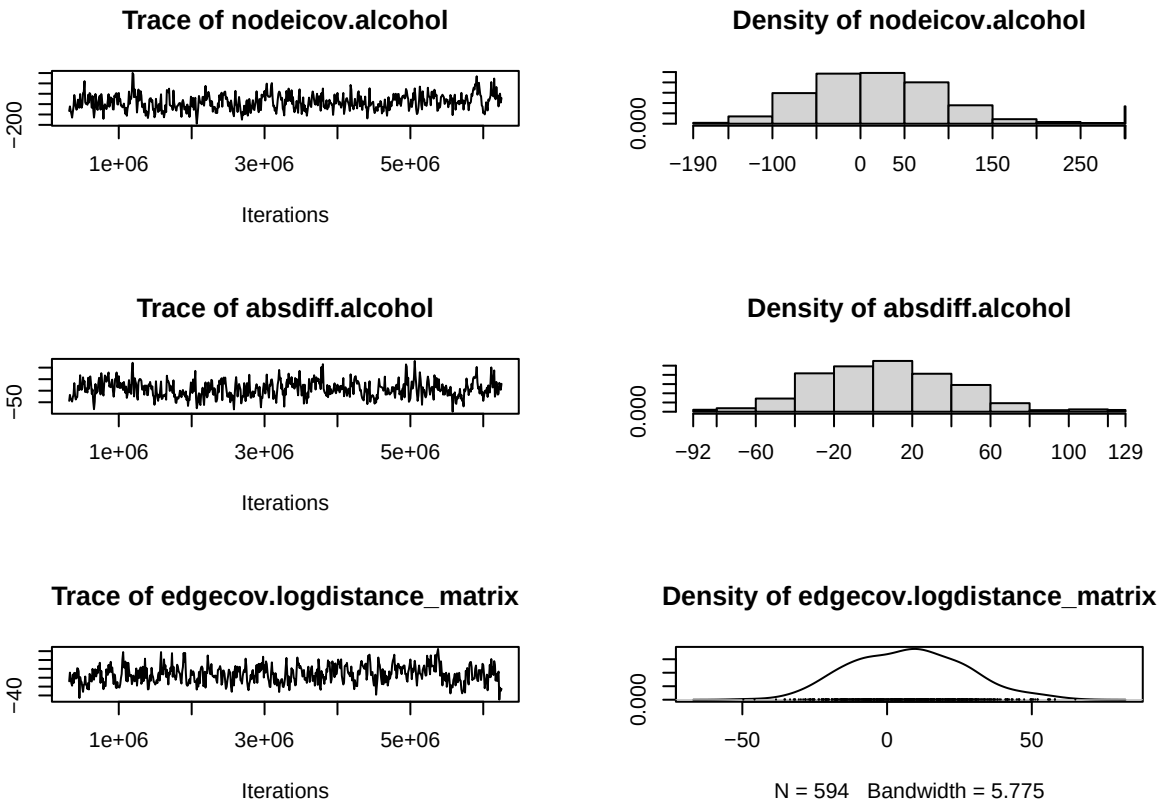
(2)

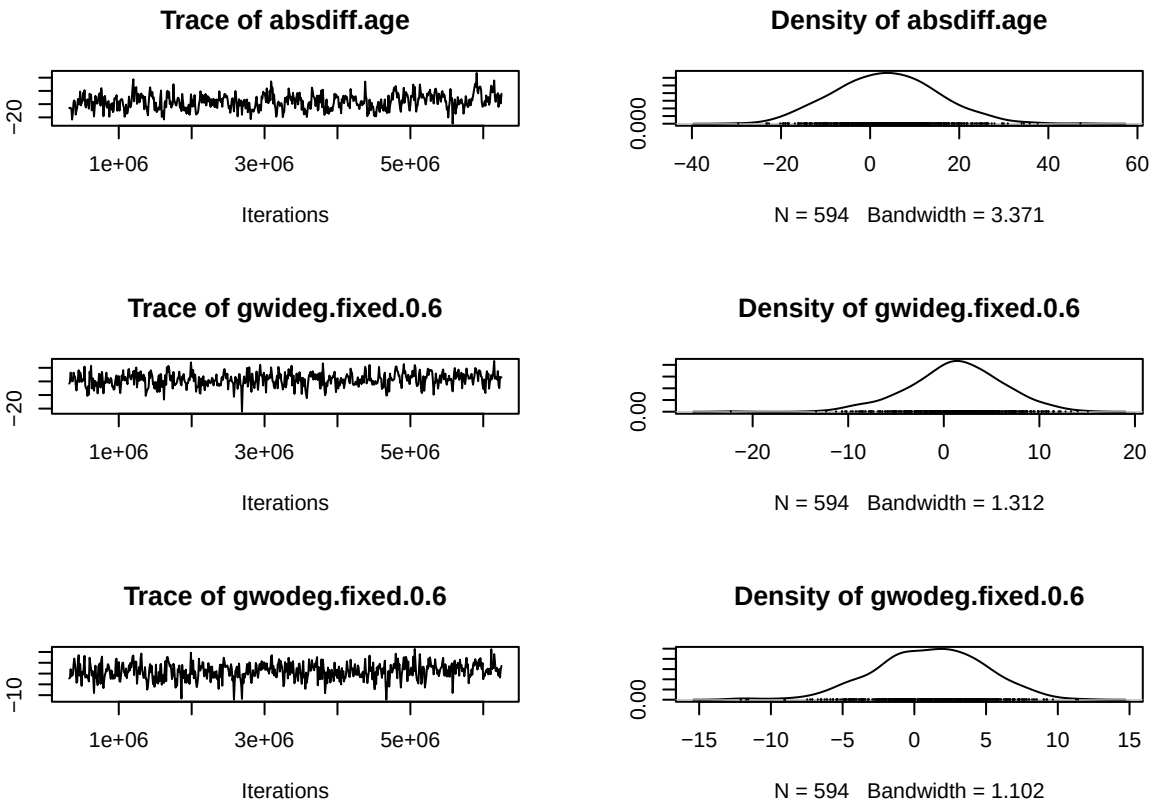
```r
ergm_model_2_params <- glasgow_net1 ~ edges +
  # Endogenous Hypotheses
  mutual +
  gwesp(0.1, fixed = TRUE) +

  # Exogenous Hypotheses: Gender
  nodeofactor("gender", levels = 1) +
  nodeifactor("gender", levels = 1) +

  # Exogenous Hypotheses: Alcohol
  nodeocov("alcohol") +
  nodeicov("alcohol") +
  absdiff("alcohol") +

  # Basic Exogenous Controls
  edgecov(logdistance_matrix) +
  absdiff("age") +
```

```r
  # Basic Endogenous Controls (Stabilizers)
  gwidegree(0.6, fixed = TRUE) +
  gwodegree(0.6, fixed = TRUE)
noisy_model_2 <- ergm(
                ergm_model_2_params,
                # ERGMs with this many parameters can take a moment to compute
                control = control.ergm(
                  MCMC.samplesize = 5000,
                  MCMC.interval = 8000,
                  MCMC.burnin = 1 * 1000 * 1000,
                  main.method = "Stochastic-Approximation",

                  parallel = 8,
                  parallel.type = "PSOCK"
                )
)
rough_estimates <- coef(noisy_model_2)
model_2 <- ergm(
  ergm_model_2_params,
  control = control.ergm(
    MCMC.samplesize = 5000,
    MCMC.interval = 10 * 1000,
    MCMC.burnin = 100 * 1000,
    init = rough_estimates,

    parallel = 8,
    parallel.type = "PSOCK"
  )
)
```

(3)


Trace of edges
−50
1e+06
3e+06
5e+06
Iterations
Density of edges
0.000
−81
−40
0
20
40
60
80
102
Trace of mutual
−30
1e+06
3e+06
5e+06
Iterations
Density of mutual
0.00
−29
−20
−10
0
5
15
25
35
Trace of gwesp.OTP.fixed.0.1
−50
1e+06
3e+06
5e+06
Iterations
Density of gwesp.OTP.fixed.0.1
0.000
−100
−50
0
50
100
150
N = 594 Bandwidth = 9.703
Trace of nodeofactor.gender.1
−60
1e+06
3e+06
5e+06
Iterations
Density of nodeofactor.gender.1
0.000
−71
−50
−30
−10
10
30
50
70
Trace of nodeifactor.gender.1
−60
1e+06
3e+06
5e+06
Iterations
Density of nodeifactor.gender.1
0.000
−74
−50
−30
−10
10
30
50
67
Trace of nodeocov.alcohol
−200
1e+06
3e+06
5e+06
Iterations
Density of nodeocov.alcohol
0.000
−186
−100
0
50
150
250


Trace of nodeicov.alcohol
−200
1e+06
3e+06
5e+06
Iterations
Density of nodeicov.alcohol
0.000
−190
−100
0
50
150
250
Trace of absdiff.alcohol
−50
1e+06
3e+06
5e+06
Iterations
Density of absdiff.alcohol
0.000
−92
−60
−20
20
60
100
129
Trace of edgecov.logdistance_matrix
−40
1e+06
3e+06
5e+06
Iterations
Density of edgecov.logdistance_matrix
0.000
−50
0
50
N = 594 Bandwidth = 5.775



We see that the distributions are unimodal and mostly centered around 0. The traces oscillate around 0 without any patterns.

```
## , , edges
##
##                edges       mutual gwesp.OTP.fixed.0.1 nodeofactor.gender.1
## Lag 0     1.00000000  0.816881966          0.92816968          0.865882514
## Lag 10000 0.48710537  0.513263513          0.48395189          0.436357439
## Lag 50000 0.14319691  0.108504900          0.12230601          0.146777966
## Lag 1e+05 0.03285285 -0.006316682          0.01262307          0.080795647
## Lag 5e+05 0.04624761  0.048280482          0.03937316          0.003960612
##           nodeifactor.gender.1 nodeocov.alcohol nodeicov.alcohol
## Lag 0               0.82220238       0.94651180       0.92719057
## Lag 10000           0.39801994       0.45914143       0.45321598
## Lag 50000           0.14230815       0.13965908       0.13169741
## Lag 1e+05           0.08624304       0.02906140       0.01466238
## Lag 5e+05           0.01329396       0.02557563       0.01657256
##           absdiff.alcohol edgecov.logdistance_matrix absdiff.age
## Lag 0        0.6673729481                0.215179484  0.80816077
## Lag 10000    0.2955580719                0.039587077  0.40636818
## Lag 50000    0.0931728651                0.001068699  0.15484790
## Lag 1e+05    0.0007558383               -0.012239971  0.08707392
## Lag 5e+05   -0.0442199130               -0.006084834  0.03205944
##           gwideg.fixed.0.6 gwodeg.fixed.0.6
## Lag 0         6.199444e-01      0.725245497
## Lag 10000     2.744258e-01      0.293916251
```

```
## Lag 50000     2.588010e-02      0.069416255
## Lag 1e+05    -3.204580e-02      0.035645903
## Lag 5e+05    -5.048898e-05      0.008934868
##
## , , mutual
##
##                edges      mutual gwesp.OTP.fixed.0.1 nodeofactor.gender.1
## Lag 0     0.81688197  1.00000000          0.88551722           0.71093383
## Lag 10000 0.48617867  0.56697842          0.50540268           0.41620936
## Lag 50000 0.15252394  0.10935759          0.13366024           0.13368238
## Lag 1e+05 0.00408791 -0.05081787         -0.02825670           0.04555040
## Lag 5e+05 0.01956479  0.03222615          0.01611993          -0.02536233
##           nodeifactor.gender.1 nodeocov.alcohol nodeicov.alcohol
## Lag 0               0.68488543       0.78213504      0.738012179
## Lag 10000           0.39905292       0.46272527      0.441980515
## Lag 50000           0.14468809       0.14350516      0.144531537
## Lag 1e+05           0.05712988       0.01103837      0.005076107
## Lag 5e+05          -0.03659235       0.01347185      0.010384009
##           absdiff.alcohol edgecov.logdistance_matrix absdiff.age
## Lag 0          0.51344846                0.166477125  0.64643430
## Lag 10000      0.29698735                0.070596635  0.38699847
## Lag 50000      0.10589352               -0.033438785  0.16749160
## Lag 1e+05      0.02157138               -0.019207675  0.04468801
## Lag 5e+05     -0.05181834               -0.003774826  0.02962123
##           gwideg.fixed.0.6 gwodeg.fixed.0.6
## Lag 0          0.592702921      0.626285308
## Lag 10000      0.393197883      0.360138449
## Lag 50000      0.060251575      0.079129264
## Lag 1e+05     -0.036448643      0.003821093
## Lag 5e+05      0.001254759      0.018511916
##
## , , gwesp.OTP.fixed.0.1
##
##                edges      mutual gwesp.OTP.fixed.0.1 nodeofactor.gender.1
## Lag 0     0.92816968  0.88551722        1.0000000000          0.812606316
## Lag 10000 0.48978210  0.54272274        0.5039316834          0.424995448
## Lag 50000 0.12949650  0.08841039        0.1108052427          0.120950729
## Lag 1e+05 0.02375997 -0.01497089       -0.0009792978          0.067496946
## Lag 5e+05 0.03167822  0.03226618        0.0243407887         -0.007069604
##           nodeifactor.gender.1 nodeocov.alcohol nodeicov.alcohol
## Lag 0              0.773553708       0.88705418      0.857197536
## Lag 10000          0.401698833       0.47329602      0.461803657
## Lag 50000          0.117609093       0.12236013      0.121543775
## Lag 1e+05          0.074805129       0.02999098      0.015826926
## Lag 5e+05         -0.005303089       0.01556343      0.009819895
##           absdiff.alcohol edgecov.logdistance_matrix absdiff.age
## Lag 0        0.5953934287                0.203367436  0.74654545
## Lag 10000    0.2995411490                0.056117820  0.40881035
## Lag 50000    0.0893606218                0.009155927  0.16353434
## Lag 1e+05    0.0002565782               -0.019819745  0.07841603
## Lag 5e+05   -0.0469088517               -0.016707060  0.03769112
##           gwideg.fixed.0.6 gwodeg.fixed.0.6
## Lag 0           0.55447539     0.6526390120
## Lag 10000       0.32223117     0.3397850486
```

```
## Lag 50000       0.02341886     0.0607661854
## Lag 1e+05      -0.02018764     0.0398037124
## Lag 5e+05      -0.02608973    -0.0008607527
##
## , , nodeofactor.gender.1
##
##                edges      mutual gwesp.OTP.fixed.0.1 nodeofactor.gender.1
## Lag 0     0.86588251  0.71093383          0.81260632          1.000000000
## Lag 10000 0.43687146  0.45618099          0.43469739          0.493281897
## Lag 50000 0.12430991  0.08699096          0.10144832          0.154754933
## Lag 1e+05 0.04258968 -0.01291907          0.01997273          0.065371776
## Lag 5e+05 0.03127289  0.04259604          0.02778696          0.008466177
##           nodeifactor.gender.1 nodeocov.alcohol nodeicov.alcohol
## Lag 0               0.87202163       0.78399372      0.783723900
## Lag 10000           0.47075590       0.40127888      0.388739029
## Lag 50000           0.13859169       0.09634908      0.090207328
## Lag 1e+05           0.07615021       0.03875498      0.027154977
## Lag 5e+05           0.01849005       0.01420449      0.007204906
##           absdiff.alcohol edgecov.logdistance_matrix absdiff.age
## Lag 0          0.58879936                 0.18619385  0.69231334
## Lag 10000      0.29612548                 0.04607284  0.33778685
## Lag 50000      0.11021869                 0.01692880  0.13384595
## Lag 1e+05      0.03112329                -0.04322336  0.10135181
## Lag 5e+05     -0.05833728                -0.05857996  0.01716604
##           gwideg.fixed.0.6 gwodeg.fixed.0.6
## Lag 0         0.5252529708     0.5667048226
## Lag 10000     0.2271277809     0.2423246936
## Lag 50000     0.0225758466     0.0477213721
## Lag 1e+05    -0.0652516651     0.0087980365
## Lag 5e+05     0.0009523676     0.0003484596
##
## , , nodeifactor.gender.1
##
##                edges      mutual gwesp.OTP.fixed.0.1 nodeofactor.gender.1
## Lag 0     0.82220238  0.68488543          0.77355371           0.87202163
## Lag 10000 0.40858662  0.42646735          0.40912132           0.49049651
## Lag 50000 0.08704235  0.03466908          0.06357052           0.11081593
## Lag 1e+05 0.01949501 -0.03325628         -0.01411577           0.04509988
## Lag 5e+05 0.07174598  0.09730161          0.07430182           0.03329906
##           nodeifactor.gender.1 nodeocov.alcohol nodeicov.alcohol
## Lag 0               1.00000000       0.75152456      0.706254951
## Lag 10000           0.46940972       0.36561236      0.360287876
## Lag 50000           0.09188772       0.07212343      0.065885064
## Lag 1e+05           0.04663674       0.01528805      0.007807263
## Lag 5e+05           0.03633832       0.04493730      0.038828342
##           absdiff.alcohol edgecov.logdistance_matrix absdiff.age
## Lag 0         0.565907519                 0.13732788  0.67213524
## Lag 10000     0.290944610                 0.02347236  0.34138041
## Lag 50000     0.104120281                -0.01515129  0.08639839
## Lag 1e+05     0.001650277                -0.02220314  0.07905838
## Lag 5e+05    -0.038687915                -0.02076841  0.05672317
##           gwideg.fixed.0.6 gwodeg.fixed.0.6
## Lag 0           0.50062029     0.5708514509
## Lag 10000       0.22740464     0.2171546398
```

```
## Lag 50000      -0.01753014     0.0177349532
## Lag 1e+05      -0.06156907     0.0008182683
## Lag 5e+05       0.04396335     0.0577969043
##
## , , nodeocov.alcohol
##
##                edges     mutual gwesp.OTP.fixed.0.1 nodeofactor.gender.1
## Lag 0     0.94651180 0.78213504          0.88705418           0.78399372
## Lag 10000 0.47232277 0.50064419          0.47645447           0.40004588
## Lag 50000 0.15017893 0.12029990          0.13508575           0.14444060
## Lag 1e+05 0.04535002 0.01422172          0.03238740           0.11029737
## Lag 5e+05 0.01345769 0.01343371          0.01035496          -0.02413801
##           nodeifactor.gender.1 nodeocov.alcohol nodeicov.alcohol
## Lag 0              0.751524563      1.000000000       0.95651518
## Lag 10000          0.367633386      0.496349279       0.50068821
## Lag 50000          0.140770734      0.158435065       0.15186315
## Lag 1e+05          0.106930340      0.034099352       0.01894001
## Lag 5e+05         -0.002603373     -0.008885438      -0.01751141
##           absdiff.alcohol edgecov.logdistance_matrix absdiff.age
## Lag 0         0.664641606                0.186109510 0.752086639
## Lag 10000     0.306274378                0.029697746 0.394583290
## Lag 50000     0.084841690                0.002264564 0.173738439
## Lag 1e+05     0.006700389               -0.011848868 0.104339701
## Lag 5e+05    -0.048671562                0.005912881 0.008702356
##           gwideg.fixed.0.6 gwodeg.fixed.0.6
## Lag 0          0.558387682       0.71774904
## Lag 10000      0.258924497       0.29612734
## Lag 50000      0.029597012       0.09302696
## Lag 1e+05     -0.006212009       0.04603186
## Lag 5e+05     -0.033652164      -0.01296483
##
## , , nodeicov.alcohol
##
##                edges      mutual gwesp.OTP.fixed.0.1 nodeofactor.gender.1
## Lag 0     0.92719057 0.738012179         0.857197536          0.783723900
## Lag 10000 0.45129644 0.473812757         0.450524300          0.383273774
## Lag 50000 0.13856007 0.114449759         0.120160641          0.136243236
## Lag 1e+05 0.05027724 0.024056015         0.045189376          0.111046298
## Lag 5e+05 0.01970317 0.002706348         0.009787506         -0.008432705
##           nodeifactor.gender.1 nodeocov.alcohol nodeicov.alcohol
## Lag 0              0.706254951      0.956515177       1.00000000
## Lag 10000          0.340208370      0.484555059       0.49316058
## Lag 50000          0.130087330      0.153481614       0.14728027
## Lag 1e+05          0.113639912      0.039336945       0.02363024
## Lag 5e+05          0.007860727     -0.007936678      -0.01171759
##           absdiff.alcohol edgecov.logdistance_matrix absdiff.age
## Lag 0          0.68144046               0.1653492012  0.74216374
## Lag 10000      0.28342948               0.0020195758  0.37517877
## Lag 50000      0.06749460               0.0003925599  0.16987370
## Lag 1e+05      0.01542303              -0.0271221910  0.11202930
## Lag 5e+05     -0.04080038               0.0185177857  0.00813744
##           gwideg.fixed.0.6 gwodeg.fixed.0.6
## Lag 0          0.543822560      0.694855847
## Lag 10000      0.234385820      0.292333256
```

```
## Lag 50000      0.034632955      0.087344597
## Lag 1e+05     -0.008464556      0.053551471
## Lag 5e+05     -0.017330294     -0.002377203
##
## , , absdiff.alcohol
##
##                 edges        mutual gwesp.OTP.fixed.0.1 nodeofactor.gender.1
## Lag 0      0.66737295  0.5134484612         0.595393429           0.58879936
## Lag 10000  0.28182110  0.2864899758         0.269239170           0.25081092
## Lag 50000  0.03429140 -0.0194224787         0.001321959           0.01978097
## Lag 1e+05  0.03353585 -0.0006949375         0.021090210           0.08047292
## Lag 5e+05 -0.01387448  0.0056384935        -0.013233664          -0.05439600
##           nodeifactor.gender.1 nodeocov.alcohol nodeicov.alcohol
## Lag 0             0.5659075191       0.66464161       0.68144046
## Lag 10000         0.2342528613       0.29366768       0.29013053
## Lag 50000         0.0009014279       0.04943974       0.05416012
## Lag 1e+05         0.0622087456       0.02928099       0.02756153
## Lag 5e+05        -0.0367856723      -0.03428566      -0.04314814
##           absdiff.alcohol edgecov.logdistance_matrix absdiff.age
## Lag 0          1.00000000                0.099611545  0.56666347
## Lag 10000      0.45469746               -0.016736882  0.25145891
## Lag 50000      0.12427341                0.012595712  0.05422513
## Lag 1e+05      0.08178266                0.001101376  0.08767272
## Lag 5e+05     -0.08346691               -0.020993984 -0.04323815
##           gwideg.fixed.0.6 gwodeg.fixed.0.6
## Lag 0           0.43763574      0.553161631
## Lag 10000       0.18237474      0.209967370
## Lag 50000      -0.03000947     -0.005592755
## Lag 1e+05      -0.07395936     -0.006270411
## Lag 5e+05      -0.03252068     -0.023778380
##
## , , edgecov.logdistance_matrix
##
##                 edges     mutual gwesp.OTP.fixed.0.1 nodeofactor.gender.1
## Lag 0     0.21517948 0.16647713          0.20336744          0.186193854
## Lag 10000 0.07210402 0.10544799          0.07547867          0.065470087
## Lag 50000 0.03298425 0.05836221          0.04222916          0.030925859
## Lag 1e+05 0.07112043 0.05058982          0.04814898          0.058507824
## Lag 5e+05 0.06369468 0.06354353          0.05809634         -0.001394434
##           nodeifactor.gender.1 nodeocov.alcohol nodeicov.alcohol
## Lag 0              0.137327878       0.18610951      0.165349201
## Lag 10000          0.028196895       0.04398567      0.045235295
## Lag 50000          0.041233380       0.02073831     -0.005149275
## Lag 1e+05          0.070668423       0.05884322      0.054129695
## Lag 5e+05         -0.005870487       0.06935632      0.043576077
##           absdiff.alcohol edgecov.logdistance_matrix absdiff.age
## Lag 0          0.09961155                1.000000000  0.17055878
## Lag 10000      0.04928979                0.485482542  0.08093368
## Lag 50000      0.01432417                0.041359782  0.09067085
## Lag 1e+05      0.02291998                0.001319171  0.05129368
## Lag 5e+05      0.02112061                0.068725472  0.05350230
##           gwideg.fixed.0.6 gwodeg.fixed.0.6
## Lag 0          0.220575255       0.20885073
## Lag 10000      0.117829078       0.08355628
```

```
## Lag 50000      0.040941542      -0.01768370
## Lag 1e+05      0.003581539       0.03822913
## Lag 5e+05      0.046187255       0.06035168
##
## , , absdiff.age
##
##                edges       mutual gwesp.OTP.fixed.0.1 nodeofactor.gender.1
## Lag 0     0.80816077  0.646434303         0.746545451           0.69231334
## Lag 10000 0.39326011  0.400764044         0.390082505           0.37326954
## Lag 50000 0.11182277  0.068044804         0.079364633           0.11301008
## Lag 1e+05 0.02530274 -0.005184593         0.003287555           0.07692778
## Lag 5e+05 0.04534674  0.060508318         0.053430341           0.01381816
##           nodeifactor.gender.1 nodeocov.alcohol nodeicov.alcohol
## Lag 0               0.67213524       0.75208664       0.74216374
## Lag 10000           0.32144208       0.35251059       0.35916141
## Lag 50000           0.10851766       0.09851711       0.08423083
## Lag 1e+05           0.07547327       0.01909821       0.02010760
## Lag 5e+05           0.02312745       0.03024495       0.03003396
##           absdiff.alcohol edgecov.logdistance_matrix absdiff.age
## Lag 0          0.56666347                 0.17055878  1.00000000
## Lag 10000      0.26234681                 0.06865364  0.49951417
## Lag 50000      0.07706578                 0.02255197  0.15907843
## Lag 1e+05      0.02331502                 0.02250639  0.10407082
## Lag 5e+05     -0.07817025                -0.02617942 -0.02390874
##           gwideg.fixed.0.6 gwodeg.fixed.0.6
## Lag 0          0.498928441       0.59967043
## Lag 10000      0.222530450       0.24157718
## Lag 50000      0.013665047       0.03411102
## Lag 1e+05     -0.002494431       0.02542859
## Lag 5e+05      0.037798732       0.02612716
##
## , , gwideg.fixed.0.6
##
##                edges     mutual gwesp.OTP.fixed.0.1 nodeofactor.gender.1
## Lag 0     0.61994439 0.59270292         0.554475388           0.52525297
## Lag 10000 0.26307455 0.37973341         0.303530611           0.26099236
## Lag 50000 0.05037000 0.03001152         0.043392271           0.04481248
## Lag 1e+05 0.00894132 0.02573584         0.002835178           0.03221592
## Lag 5e+05 0.04248147 0.06707443         0.053494060           0.00202563
##           nodeifactor.gender.1 nodeocov.alcohol nodeicov.alcohol
## Lag 0              0.500620292       0.55838768       0.54382256
## Lag 10000          0.247063133       0.22892153       0.21964807
## Lag 50000          0.051528572       0.06471339       0.05308093
## Lag 1e+05          0.024784947       0.02959999       0.02768090
## Lag 5e+05         -0.005320667       0.04686984       0.03845467
##           absdiff.alcohol edgecov.logdistance_matrix absdiff.age
## Lag 0         0.437635742                 0.220575255  0.49892844
## Lag 10000     0.144151351                 0.069309986  0.17267495
## Lag 50000     0.040422378                -0.015497288  0.03733483
## Lag 1e+05    -0.002583239                -0.050313136  0.02088915
## Lag 5e+05    -0.025814752                 0.002130298  0.01437920
##           gwideg.fixed.0.6 gwodeg.fixed.0.6
## Lag 0           1.00000000       0.67072488
## Lag 10000       0.33671182       0.26961615
```

```
## Lag 50000      -0.01368310       0.01845899
## Lag 1e+05      -0.04229845       0.01507552
## Lag 5e+05       0.01184779       0.05299375
##
## , , gwodeg.fixed.0.6
##
##                edges      mutual gwesp.OTP.fixed.0.1 nodeofactor.gender.1
## Lag 0     0.72524550 0.626285308          0.65263901           0.56670482
## Lag 10000 0.30320122 0.391805545          0.33705756           0.27265279
## Lag 50000 0.05675826 0.007443771          0.04074714           0.06424461
## Lag 1e+05 0.04080408 0.007522402          0.02289428           0.10106897
## Lag 5e+05 0.08360301 0.065313268          0.06882107           0.06334969
##           nodeifactor.gender.1 nodeocov.alcohol nodeicov.alcohol
## Lag 0               0.57085145       0.71774904       0.69485585
## Lag 10000           0.22368062       0.29303832       0.30589704
## Lag 50000           0.06934962       0.05754199       0.05878202
## Lag 1e+05           0.07712974       0.05202599       0.04838083
## Lag 5e+05           0.08021381       0.04771619       0.04289793
##           absdiff.alcohol edgecov.logdistance_matrix absdiff.age
## Lag 0          0.55316163                 0.20885073  0.59967043
## Lag 10000      0.19172024                 0.03170473  0.21902021
## Lag 50000      0.01079418                -0.05535068  0.07517014
## Lag 1e+05      0.02251485                -0.04797461  0.05914249
## Lag 5e+05      0.01137403                 0.06185459  0.06472791
##           gwideg.fixed.0.6 gwodeg.fixed.0.6
## Lag 0           0.67072488       1.00000000
## Lag 10000       0.27744753       0.26183701
## Lag 50000      -0.03656495       0.04026944
## Lag 1e+05      -0.01447244       0.03976477
## Lag 5e+05       0.04743393       0.02617803
```

From the lags we see that the model explores the state space very well, since autocorrelation becomes very close to 0.

(4)

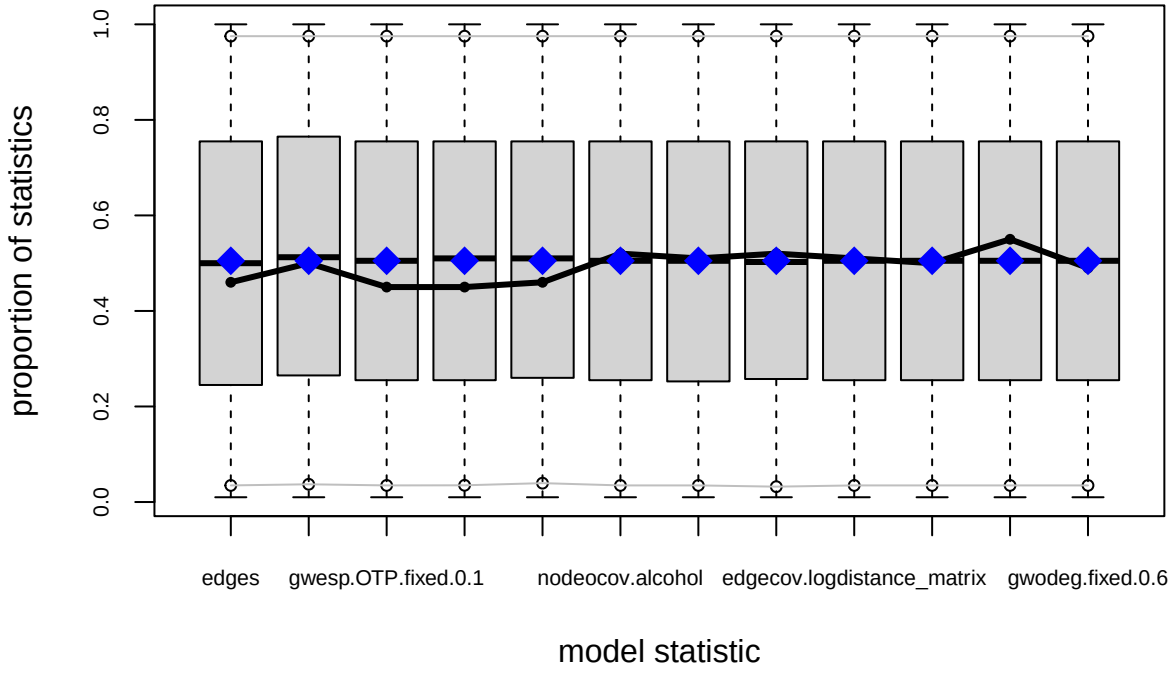
```r
model_2_gof <-  gof(
  model_2,
  GOF = ~ idegree + odegree + espartners + distance
)
```

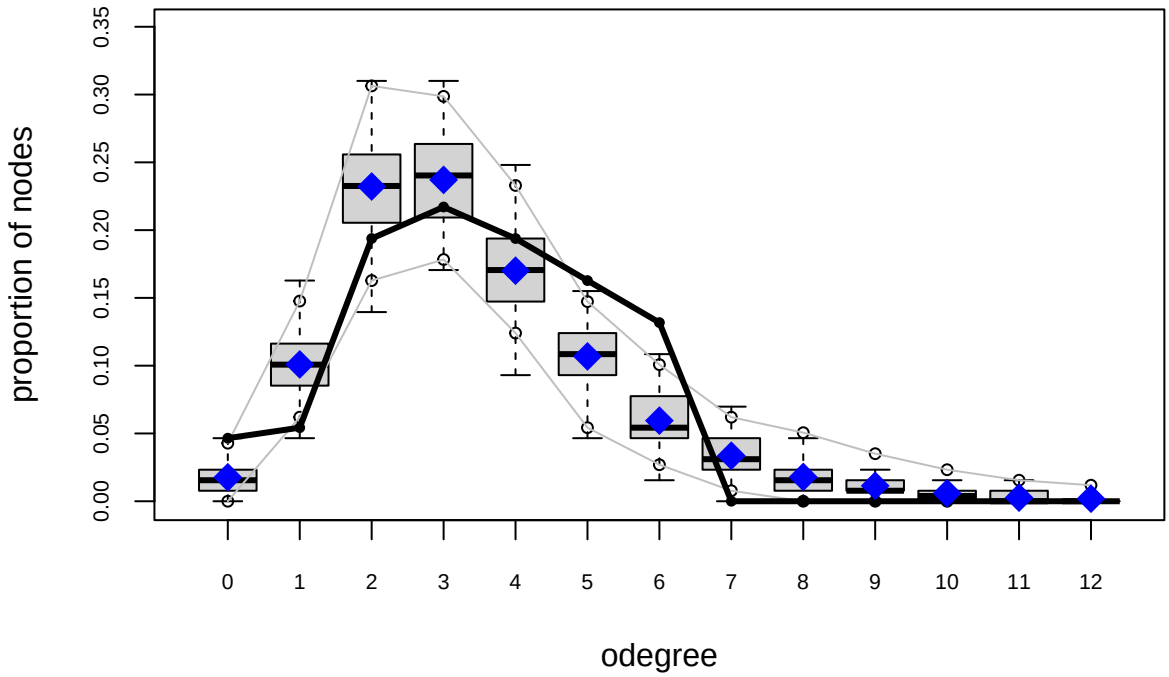
```
##
## Goodness-of-fit for model statistics
##
##                                  obs      min      mean       max MC p-value
## edges                        448.0000 371.0000  448.4400  536.0000       0.92
## mutual                       124.0000  95.0000  123.9800  161.0000       1.00
## gwesp.OTP.fixed.0.1          325.1830 237.1026  325.5593  441.7686       0.90
## nodeofactor.gender.1         266.0000 211.0000  264.2000  327.0000       0.90
## nodeifactor.gender.1         266.0000 207.0000  265.2200  328.0000       0.92
## nodeocov.alcohol            1084.0000 891.0000 1087.0100 1268.0000       0.96
```

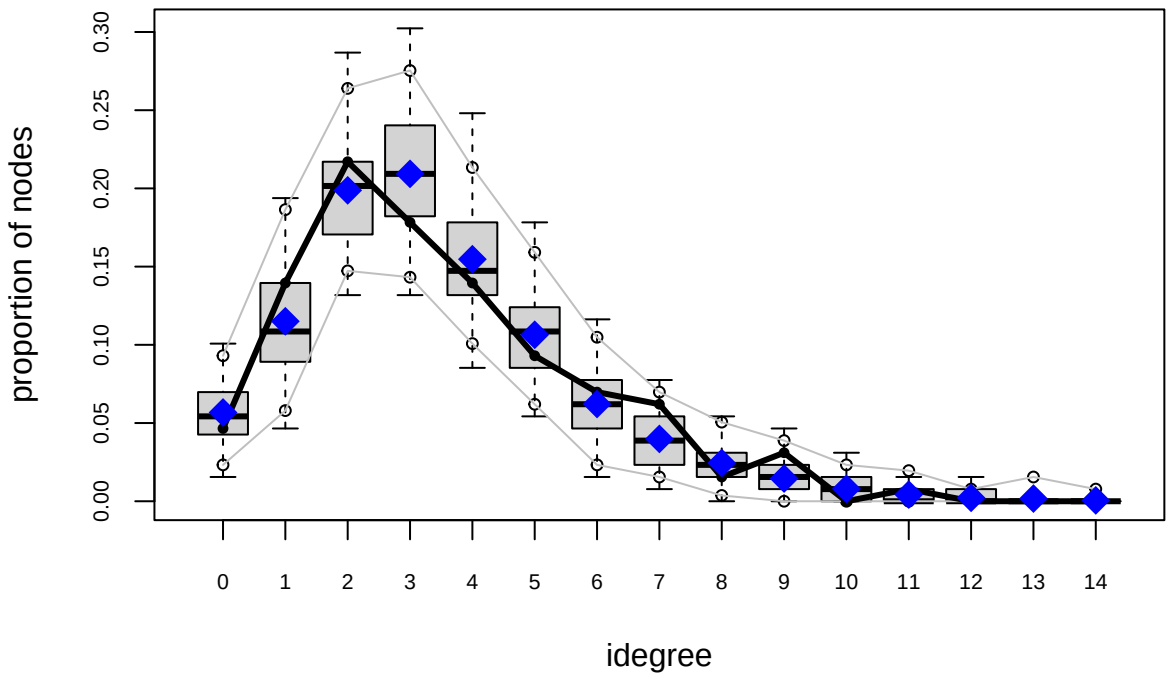
```
## nodeicov.alcohol           1136.0000 933.0000 1139.7400 1325.0000       0.98
## absdiff.alcohol             446.0000 355.0000  445.5700  520.0000       0.96
## edgecov.logdistance_matrix   97.1500  55.2600   97.5394  140.0200       0.98
## absdiff.age                 146.0000 125.1000  146.5960  181.5000       1.00
## gwideg.fixed.0.6            193.1131 181.1744  193.3385  207.1563       0.90
## gwodeg.fixed.0.6            201.5419 192.8048  201.6836  210.0497       0.98
##
## Goodness-of-fit for minimum geodesic distance
##
##      obs  min    mean  max MC p-value
## 1    448  371  448.44  536       0.92
## 2    777  739 1185.92 1879       0.02
## 3   1162 1325 2507.24 4208       0.00
## 4   1580 1877 3638.99 4797       0.00
## 5   1802 2066 3348.41 4186       0.00
## 6   1739 1067 2061.30 2858       0.42
## 7   1625  261  974.03 1991       0.12
## 8   1274   42  389.91 1280       0.02
## 9    872    0  145.12  729       0.00
## 10   573    0   49.75  488       0.00
## 11   287    0   18.09  365       0.02
## 12   103    0    7.14  289       0.02
## 13    21    0    3.61  244       0.06
## 14     5    0    1.82  157       0.06
## 15     0    0    1.27  122       1.00
## 16     0    0    0.67   67       1.00
## 17     0    0    0.53   53       1.00
## 18     0    0    0.17   17       1.00
## 19     0    0    0.14   14       1.00
## Inf 4244  635 1729.45 3360       0.00
##
## Goodness-of-fit for out-degree
##
##    obs min  mean max MC p-value
## 0    6   0  2.28   6       0.06
## 1    7   4 13.02  21       0.04
## 2   25  18 29.95  45       0.36
## 3   28  22 30.58  40       0.68
## 4   25  12 21.94  32       0.56
## 5   21   6 13.78  24       0.02
## 6   17   2  7.67  14       0.00
## 7    0   0  4.33   9       0.02
## 8    0   0  2.31   8       0.18
## 9    0   0  1.47   5       0.46
## 10   0   0  0.76   5       1.00
## 11   0   0  0.34   3       1.00
## 12   0   0  0.24   2       1.00
## 13   0   0  0.12   1       1.00
## 14   0   0  0.07   1       1.00
## 15   0   0  0.06   1       1.00
## 16   0   0  0.05   1       1.00
## 18   0   0  0.01   1       1.00
## 20   0   0  0.01   1       1.00
## 24   0   0  0.01   1       1.00
```

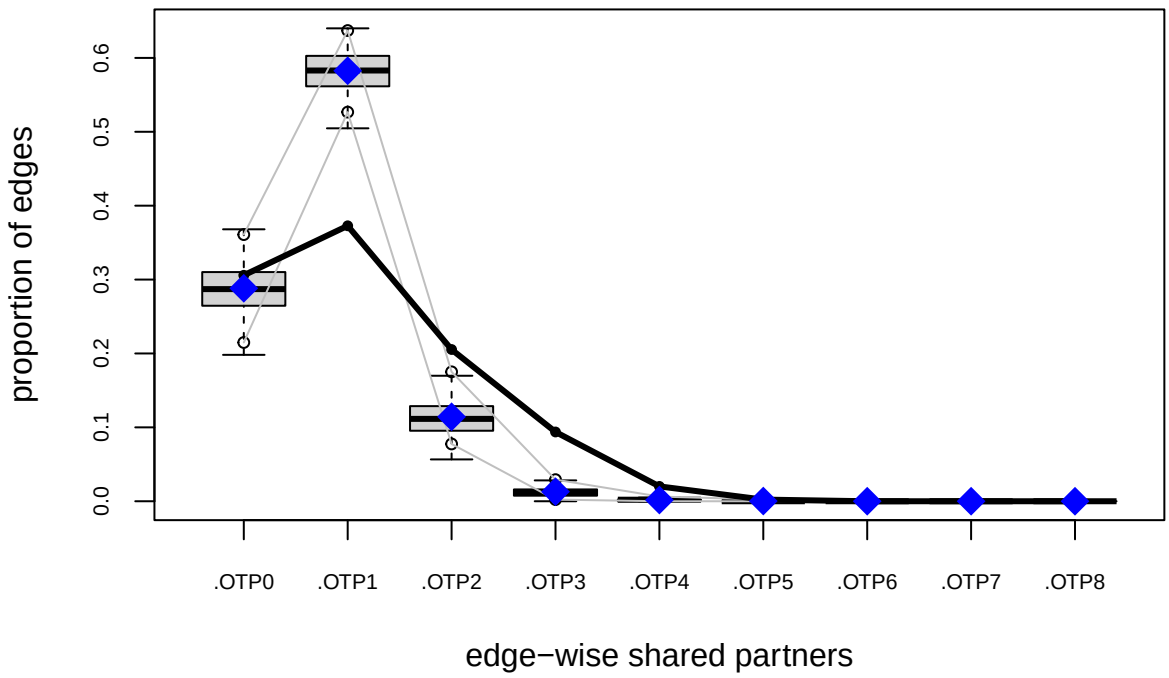
```
##
## Goodness-of-fit for in-degree
##
##    obs min  mean max MC p-value
## 0    6   2  7.32  13       0.78
## 1   18   6 14.84  28       0.52
## 2   28  17 25.64  38       0.62
## 3   23  17 27.00  39       0.50
## 4   18  11 19.96  32       0.82
## 5   12   7 13.73  23       0.76
## 6    9   2  8.02  15       0.78
## 7    8   1  5.14  10       0.36
## 8    2   0  3.15   7       0.78
## 9    4   0  1.87   7       0.18
## 10   0   0  1.03   4       0.70
## 11   1   0  0.55   3       0.84
## 12   0   0  0.28   2       1.00
## 13   0   0  0.22   2       1.00
## 14   0   0  0.07   1       1.00
## 15   0   0  0.04   1       1.00
## 16   0   0  0.07   1       1.00
## 17   0   0  0.03   1       1.00
## 20   0   0  0.01   1       1.00
## 23   0   0  0.02   1       1.00
## 26   0   0  0.01   1       1.00
##
## Goodness-of-fit for edgewise shared partner
##
##       obs min   mean max MC p-value
## .OTP0 137 103 128.51 157       0.38
## .OTP1 167 198 261.43 321       0.00
## .OTP2  92  21  51.71 103       0.02
## .OTP3  42   0   5.91  16       0.00
## .OTP4   9   0   0.74   4       0.00
## .OTP5   1   0   0.11   2       0.20
## .OTP6   0   0   0.01   1       1.00
## .OTP7   0   0   0.02   1       1.00
```

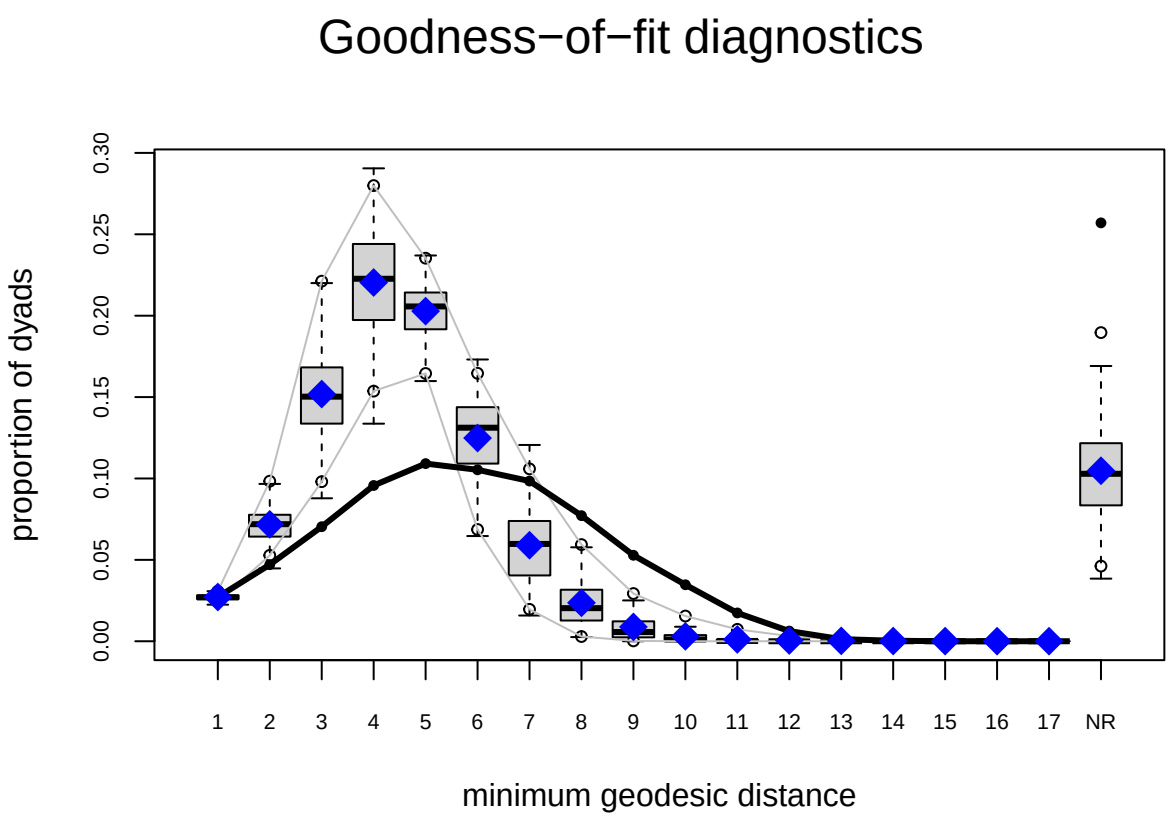
From the summary table we see that the p-values are high (not all of them are significant on the 5%) level. This also indicates that the model fits well based on the statistics that it is trained on.


proportion of statistics
0.0
0.2
0.4
0.6
0.8
1.0
edges
gwesp.OTP.fixed.0.1
nodeocov.alcohol
edgecov.logdistance_matrix
gwodeg.fixed.0.6
model statistic


proportion of nodes
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0
1
2
3
4
5
6
7
8
9
10
11
12
odegree


proportion of nodes
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
idegree


proportion of edges
0.0
0.1
0.2
0.3
0.4
0.5
0.6
.OTP0
.OTP1
.OTP2
.OTP3
.OTP4
.OTP5
.OTP6
.OTP7
.OTP8
edge−wise shared partners



Plotting the auxiliary statistics shows us that the model explores networks with similar in- and out-degree distributions, but the explored networks are much less connected than the reference one. This can be seen in the `distance` and `espartners` diagrams.

(5)

```
## Call:
## ergm(formula = ergm_model_2_params, control = control.ergm(MCMC.samplesize = 5000,
##     MCMC.interval = 10 * 1000, MCMC.burnin = 100 * 1000, init = rough_estimates,
##     parallel = 8, parallel.type = "PSOCK"))
##
## Monte Carlo Maximum Likelihood Results:
##
##                              Estimate Std. Error MCMC % z value Pr(>|z|)
## edges                        -5.66637    0.18797      0 -30.145  < 1e-04 ***
## mutual                        2.96419    0.19884      0  14.907  < 1e-04 ***
## gwesp.OTP.fixed.0.1           1.81902    0.10183      0  17.863  < 1e-04 ***
## nodeofactor.gender.1          0.01208    0.12834      0   0.094 0.925009
## nodeifactor.gender.1          0.09565    0.11378      0   0.841 0.400540
## nodeocov.alcohol             -0.12401    0.06006      0  -2.065 0.038938 *
## nodeicov.alcohol              0.15190    0.05392      0   2.817 0.004842 **
## absdiff.alcohol              -0.13989    0.03784      0  -3.697 0.000219 ***
## edgecov.logdistance_matrix -0.19396    0.05314      0  -3.650 0.000262 ***
## absdiff.age                  -0.15035    0.14784      0  -1.017 0.309179
## gwideg.fixed.0.6              0.99343    0.31039      0   3.201 0.001372 **
```

```
## gwodeg.fixed.0.6            2.30624    0.42969      0   5.367  < 1e-04 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      Null Deviance: 22890  on 16512  degrees of freedom
##  Residual Deviance:  2930  on 16500  degrees of freedom
##
## AIC: 2954  BIC: 3046  (Smaller is better. MC Std. Err. = 1.136)
```

- i - reciprocity is supported by data
- ii - transitivity (gwesp.OTP.fixed.0.1) is supported by the data
- iii, iv - cannot tell as the coefficients are insignificant
- v - the opposite is supported
- vi - supported
- vii - supported (negative coefficients, so the abs difference should be small)

(6)

????

(7)

- **H1** - children with similar literacy scores tend to become friends (literacy score homophily)
- **H2** - Girls tend to receive more friend nominations than they send
- **H3 (a bit unsure about this one)** - alcohol homophily is more prevalent in boys compared to girls